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PALAEOMAGNETIC EVIDENCE FROM THE ORDOVICIAN AND SILURIAN OF NORTHWEST PENINSULAR MALAYSIA

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The Ordovician and Silurian Setul Limestone of the Langkawi Islands, northwest Peninsular Malaysia, has a mean magnetic vector of $D = 338°$, $I = 62°$ after cleaning and correction for tilt. This is equivalent to a palaeolatitude of 43°, and a palaeomagnetic pole at 46°N, 76°E. The Silurian part of the Setul limestone also shows a similar direction. The Ordovician results are equivalent to a palaeolatitude of 43°, N or S. Recent reconstructions, based on palaeontology, place Indochina and China in the northern hemisphere in the Ordovician; if this is correct, a palaeolatitide of 43° for Langkawi would imply that Malaya-Indochina was the most northerly continental fragment at that time.

1. Introduction

Sampling for palaeomagnetic work was done in the Langkawi Islands northwest Peninsular Malaysia in 1974 and 1977, using a 1-inch field drill. Approximately 250 samples were taken from the Palaeozoic rocks, and, of these, 37 samples from the Ordovician to Silurian Setul Limestone in the northeast part of the island gave a stable magnetic direction. The remaining samples (from the Cambrian, Ordovician, Silurian, Devonian, and Carboniferous mudstone and arigillite) proved to be unsuitable, because of magnetic instability and weak magnetization.

The Ordovician limestones came from seven sites listed in Table 1. Limestones from sites A to D, and F, are grey, those from E and G (east side of Pulau Tanjung Dendang) are reddish; all are crossed by anastomosing shaley dolomitic bands, a millimetre or so thick, approximately parallel to the bedding, probably stylolitic, though Wong [1] considers them original. The bedding is clear in all the sites sampled, especially at sites E, F and G, where the sequence includes a bed of chamosite colite, about 0.5 m thick, which, with the red colour and nodular appearance of the limestone (resembling shrinkage cracks on exposed bedding surfaces) may indicate a shallow marine environment.

The Setul Limestone in Langkawi is known to range from Ordovician to Silurian [2–4]. Site D is the lowest in the succession, and lies in the lower part of the Lower to Middle Ordovician "Unit 3" of the Lower Setul Limestone [1, p. 18]; sites B and C lie within the same unit, about 400 m stratigraphically higher. Site F is at the very top of "Unit 3", some 600 m stratigraphically above B and C, whereas E and G are 5 and 7 m above the base of Wong's overlying Upper Ordovician "Unit 4". Site A lies at the base of the Upper Setul Limestone, of Silurian age, 170 m stratigraphically above G.

2. Method

The samples were measured in the laboratory of the Department of Geology, University of Malaya, on a DIGICO spinner magnetometer, and pilot specimens were demagnetized thermally step by step at 100°C intervals to 400°C, then at 50°C intervals to 600°C.

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TABLE 1

Palaeomagnetic results from the Ordovician and Silurian of Langkawi

Site	Uncorrected		Corrected for tilt		Location	Formation and age	Position strat. below site A (m)	Statistics					
									uncorrected			corrected for tilt		
	$D(°)$	$I(°)$	$D(°)$	$I(°)$				n	k	α_{95}	R	k	α_{95}	R
A	254	40	300	78	Pulau Langgun, above Lower Detrital Member; 6.45°N, 99.88°E	Upper Setul Limestone; U. Silurian	–	4	21.3	20.3	3.85977	21.3	20.3	3.85977
G	291	79	350	64	Pulau Tanjung Tanjung Dendang, east side; 6.41°N, 99.91°E	Lower Setul Limestone, "Unit 4"; U. Ordovician	170	6	62.9	8.5	5.92054	62.9	8.5	5.92054
E	320	72	350	53	same	same	172	10	173.0	3.6	9.94797	173.0	3.6	9.94797
F	336	66	340	43	grey limestone below E	Lower Setul Limestone, top of "Unit 3"; M. Ordovician	177	3	35.2	21.0	2.94327	35.2	21.0	2.94327
B	250	67	334	61	north side Kuala Ayer Hangat; 6.47°N, 99.85°E	Lower Setul Limestone, "Unit 3"; M. or L. Ordovician	800	5	47.1	11.2	4.91517	47.1	11.2	4.91517
C	255	58	313	69	same, another bay, 200 m east of B	same	800	5	121.0	6.9	4.96696	121.0	6.9	4.96696
D	257	35	320	79	Sungai Ayer Hangat; 6.44°N, 99.83°E	same	1200	4	413.1	4.5	3.99273	128.1	8.1	3.97658
Mean of site means, B to G	275	67	338	62		Lower Setul Limestone Ordovician		6	14.1	18.4	5.64630	33.3	11.7	5.85016
Mean of all samples from sites B to G	279	68	341	62		Ordovician		33	16.5	6.3	31.06910	32.3	4.4	32.01184
Palaeomagnetic pole positions, mean of poles sites B to G					latitude 46°	longitude 76°E		6	$K = 20.0$	$A_{95} = 15.3$	$\delta p = 7.6$	$\delta m = 9.8$		

D, I = declination east of true north, inclination (positive down) of cleaned remanance; R = length of resultant of n unit vectors; k, K = precision parameter; α_{95}, A_{95} = radius of circle of 95% confidence about mean. Stratigraphy from Wong [1].

Using the stability index of Briden [5], the most suitable temperature for demagnetizing the remainder of the samples was selected (200 or 300°C for these samples). The limestone samples of chamozite oolite showed a magnetization close to that of the present field, and were stable only to 300°C; these were omitted from the mean.

3. Results and discussion

Results are shown in Table 1. The inclination of 62°, equivalent to a palaeolatitude of 43° is surprising, in that the 700-m-thick limestones of the Ordovician to Silurian would appear to suggest a somewhat lower latitude. However, the timespan of the Setul Limestone is long, and it appears that these limestones did not form rapidly in the way that later limestones did.

The question may be asked as to whether the magnetism measured may be secondary, imposed on the rocks after tilting, for example by remagnetization by the intrusion of the Late Triassic granite which crops out 4–6 km away from the sites sampled; the direction of the magnetic vector before correction for tilt, however, is steeper (67°) and so a slightly higher lati-

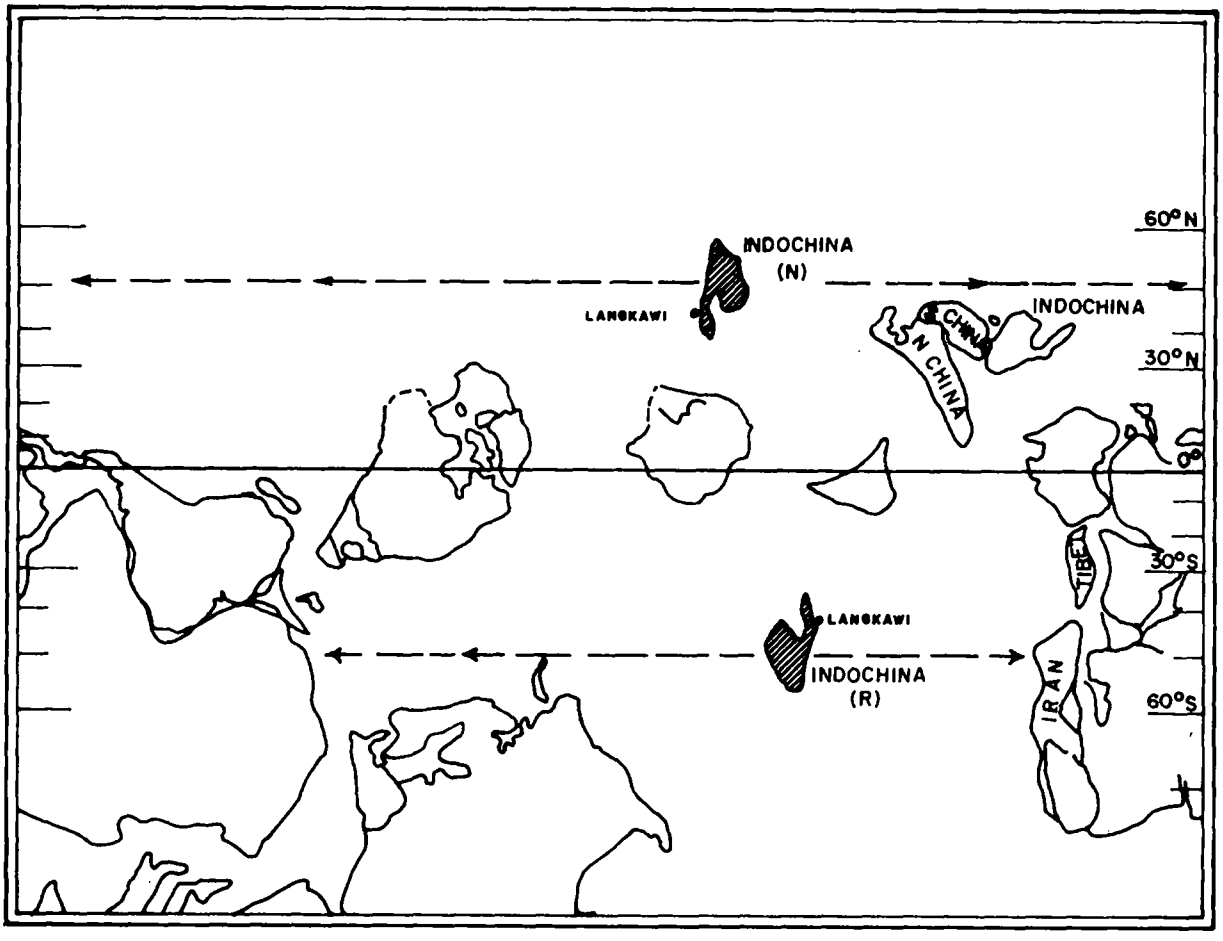


Fig. 1. World palaeogeography and position of Indochina and Malaya in Ordovician time. Unshaded areas show positions of landmasses in Middle Ordovician according to Scotese et al. [6]. Shaded masses are positions of a hypothetical Indochina-Malaya block which satisfy the data presented in this paper: N, assuming normal, R, reversed, magnetization. Present coastlines are shown for convenience of recognition. Arrows indicate that longitude of the block is not constrained.

tude would be indicated, although for a later period. The absence of reversals, and the lack of systematic change in the direction of magnetization throughout the long time period represented by the sampled sites, would seem to favour remagnetization. However, the grouping of the vector directions is considerably improved on correcting for tilt of the bedding (the precision parameter, k, doubles), suggesting that the magnetization was acquired before tilting, and stability to 400°C and more is evidence in favour of primary magnetization.

4. Conclusions

These results indicate that the Langkawi Islands (and presumably the contiguous belt of Lower Palaeozoic rocks in west Malaysia and Peninsular Thailand) lay in a latitude of 43°, north or south, in the Ordovician and Silurian. The two positions and orientations of a hypothetical Malaya-Indochina block which satisfy the palaeomagnetic data are shown in Fig. 1, superimposed on a recent reconstruction of the palaeogeography of the Middle Ordovician based on distribution of fossil faunas and palaeomagnetic evidence [6]. Palaeomagnetic evidence is sparse for the China-Indochina area, the only data available being a Late Cambrian pole for north Korea and a Middle Silurian pole for Kansu, north China [7] and some late Palaeozoic reconnaissance results for the Malay Peninsula [8]. The uncertainty in the position of China-Indochina is illustrated by the fact that an alternative reconstruction for Middle Ordovician palaeogeography shows China-Indochina across the equator and entirely south of 28°N and in a different orientation [9, fig. 3]; the authors do not comment on the discrepancy between this reconstruction and that of Scotese et al. [6, fig. 13].

Geological evidence indicates that a continental source of sediment lay west of the Malay Peninsula at times during the Palaeozoic [10]. It can be seen from Fig. 1 that the only continental blocks in the northern hemisphere in the appropriate latitude in Middle Ordovician were, according to the reconstruction of Scotese et al. [6], the north and south China blocks; according to Ziegler et al. [9], there was no land north of 30°N in the Middle Ordovician. If the Malaya-Indochina block was in the southern hemisphere a potential source of sediment could have been Iran.

The most effective way to test these possibilities would be to obtain further palaeomagnetic evidence from the Palaeozoic belt of northwest Malaya, Thailand, and southwest China. It would seem possible to construct a partial apparent polar wandering curve from the Ordovician-Silurian by more intensive sampling of the Setul Limestone. A more conclusive fold test applied by sampling the limestone in places where it dips divergently could settle the question as to whether the remanent magnetism is primary.

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